Cardiovascular Disease Prediction Through Data Preprocessing and Classification Techniques (April 2026)

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***Abstract*—Cardiovascular disease is a leading cause of mortality worldwide, and early detection remains a critical challenge in healthcare. This study predicts cardiovascular disease using exploratory data analysis, preprocessing, dimensionality reduction, and supervised classification models. Logistic Regression, K-Nearest Neighbors, Random Forest, Support Vector Machine, and a Multilayer Perceptron were trained and evaluated under a consistent validation protocol. ROC-AUC was used as the official model-selection metric, while recall, F2-score, and confusion-matrix errors were analyzed due to the clinical relevance of false negatives. The Multilayer Perceptron obtained the best validation ROC-AUC and was selected as the final model. Threshold adjustment improved sensitivity-oriented behavior for screening. PCA reduced the preprocessed representation from 16 to 9 components with minimal loss in ROC-AUC, whereas UMAP provided stronger dimensionality reduction but degraded predictive performance. These results show that careful preprocessing, model selection, and threshold tuning are essential for developing useful cardiovascular disease screening models.**

***Impact Statement*—Cardiovascular disease is a major global health challenge, accounting for millions of deaths annually. Early detection and prevention are critical to reducing this burden. This study explores the use of exploratory data analysis, preprocessing techniques, and machine learning classification models to predict cardiovascular disease based on clinical and lifestyle variables. The impact of this work lies in demonstrating how careful data preparation and model selection can enhance predictive accuracy.**

***Index Terms*—Cardiovascular disease, Classification, Data preprocessing, Exploratory data analysis, Machine learning, Predictive modeling.**

I. INTRODUCTION

THIS project addresses the prediction of cardiovascular disease using patient data. The dataset includes clinical and lifestyle variables such as age, blood pressure, cholesterol, glucose levels, smoking status, alcohol intake, physical activity, and BMI. Exploratory data analysis (EDA) revealed that the target variable is nearly balanced, with similar proportions of positive and negative cases. However, several categorical predictors are highly skewed, and crosstab analysis showed that higher cholesterol, glucose, and BMI are strongly associated with cardiovascular disease, while gender and smoking/alcohol intake exhibit minimal differences. These findings highlight the need for preprocessing techniques such as normalization, categorical encoding, handling of missing data, and careful treatment of skewed categorical distributions. Given that the target variable is binary, supervised learning in the form of classification was selected as the appropriate paradigm. This choice is justified by the availability of labeled data and the clinical relevance of metrics such as recall and F1-score. Several models were evaluated, including Logistic Regression, KNN, Random Forest, SVM, and MLP, to determine the most effective approach for predictive healthcare applications.

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II. PROBLEM DESCRIPTION

Cardiovascular disease (CVD) remains one of the leading causes of mortality worldwide. Early detection and prevention are critical to reducing morbidity and mortality rates. Traditional diagnostic methods often rely on clinical tests that are costly and time-consuming, limiting their applicability for large-scale preventive strategies. In this context, machine learning (ML) offers an opportunity to leverage patient data to build predictive models that can support clinical decision-making and promote preventive healthcare.

A. Data Base Composition

The dataset comprises approximately 70,000 samples, with a binary target variable indicating the presence or absence of cardiovascular disease. The target variable is nearly balanced, with similar proportions of positive and negative cases. Crosstab analysis revealed that higher cholesterol and glucose levels, as well as increased BMI, are strongly associated with cardiovascular disease, while gender and smoking/alcohol intake show minimal differences. Physical activity appears to have a protective effect, with more active patients showing lower prevalence of disease. Several categorical predictors are highly skewed, and issues such as missing values and outliers in blood pressure and weight highlight the importance of preprocessing.

B. Selected ML Paradigm

Given the binary nature of the target variable and the availability of labeled data, supervised learning in the form of classification was selected as the appropriate paradigm. This choice is justified by the clinical relevance of distinguishing between patients with and without cardiovascular disease, and by the suitability of classification algorithms to model discrete outcomes. Although the target distribution is balanced, skewed predictors require careful preprocessing and feature engineering.

III. State of the Art

A. Individual Studies

Study 1 — Mridha et al. (ICDAI 2023) This work used the same Kaggle Cardiovascular Disease dataset and applied a supervised classification paradigm. Techniques included Logistic Regression, Random Forest, Gradient Boosting, and SVM, complemented with SHAP and LIME for explainability. Validation was performed using k-fold cross-validation. Metrics included Accuracy, Precision, Recall, F1-score, and ROC-AUC. Results showed that ensemble methods achieved the highest predictive performance (AUC > 0.85), while explainability analysis highlighted cholesterol, glucose, and BMI as key predictors [1].

Study 2 — Al-Makhadmeh & Tolba (IEEE Access, 2020) Al-Makhadmeh and Tolba [2] proposed a supervised learning paradigm for cardiovascular disease prediction, focusing on deep learning architectures. The authors implemented a hybrid CNN–LSTM model to capture both spatial and temporal patterns in patient data. Validation was performed using k-fold cross-validation to ensure robustness. The evaluation metrics included Accuracy, Sensitivity, Specificity, and ROC-AUC, with the hybrid CNN–LSTM consistently outperforming traditional classifiers such as logistic regression and decision trees. Reported results showed accuracy above 90% and AUC values exceeding 0.90, highlighting the potential of deep learning models in medical prediction tasks.

Study 3 — Muhammad et al. (IEEE Access, 2023) Muhammad et al. proposed a machine learning framework for ischemic cardiovascular disease prediction using a Kaggle dataset with 918 observations and 12 features. The study compared several supervised learning algorithms, including K-Nearest Neighbors (KNN), Random Forest (RF), Logistic Regression (LR), Support Vector Machine (SVM), Gaussian Naive Bayes (GNB), and Decision Trees (DT). The models were evaluated using accuracy, precision, recall, F1-score, and AUC. Their best-performing model was KNN, achieving 91.8% accuracy, 92.5% precision, 91.4% recall, 91.9% F1-score, and an AUC of 90.27%. This work is relevant to the present project because it includes several of the same model families evaluated in this study, particularly KNN, RF, LR, and SVM.

Study 4 — Al Gharib et al. (IEEE Access, 2025) Al Gharib et al. introduced a hybrid explainable learning framework for cardiovascular disease detection. Their approach combined Random Forest, Gradient Boosting, and SVM as base learners, with XGBoost used as a meta-learner. The model also included polynomial feature expansion and mutual information-based feature selection. The proposed method was evaluated on three public datasets and reported accuracy values between 88% and 91%, F1-scores between 0.88 and 0.91, and AUC values up to 0.96. Explainability methods such as SHAP and LIME were used to identify relevant predictors, including cholesterol, exercise-induced angina, and age. This study is relevant because it highlights the value of ensemble methods, feature selection, and explainability in cardiovascular disease prediction.

Table I summarizes four representative studies related to cardiovascular disease prediction. The selected works cover classical supervised learning models, ensemble methods, deep learning approaches, and explainable machine learning. These studies are relevant because they report metrics such as accuracy, recall, F1-score, and ROC-AUC, which are also used in the present work.

IV. Experimental Setup and Model Evaluation

A. Experimental Protocol

The dataset was split into training, validation, and test subsets using stratified sampling to preserve the class distribution. The training set was used for model fitting and cross-validation-based hyperparameter search. The validation set was used for model comparison and threshold adjustment. The test set was held out and used only once for final performance estimation.

Cross-validation was used during hyperparameter selection to estimate the stability of each candidate configuration. The final model comparison was performed on the independent validation and test sets. When available, the variability across cross-validation folds was summarized using an approximate 95% confidence interval for ROC-AUC.

B. Preprocessing Pipeline

The preprocessing stage was designed to ensure that all models received a consistent and reproducible input representation. After the initial data cleaning process, the patient identifier was removed because it does not contain predictive clinical information. Age was converted from days to years, and body mass index (BMI) was derived from height and weight. Unrealistic anthropometric and blood pressure values were removed according to clinically plausible ranges.

Numerical variables were standardized when required by the model, particularly for distance-based and margin-based classifiers such as KNN, SVM, and MLP. Ordinal clinical variables such as cholesterol, glucose, and BMI category were encoded preserving their natural ordering. Binary lifestyle variables were kept as binary indicators. All preprocessing operations were included inside model pipelines to avoid data leakage between training, validation, and test partitions.

C. Performance Metrics

ROC-AUC was used as the official model-selection metric because it evaluates the global discriminative capacity of the classifier independently of a fixed threshold. However, because false negatives are clinically relevant in cardiovascular disease screening, F2-score, recall, and the number of false negatives were also analyzed as secondary criteria.

TABLE I
COMPARISON OF METHODS, METRICS, AND RESULTS IN SELECTED STUDIES

Study	Methods	Metrics	Main results
Mridha et al. (ICDAI 2023)	LR, RF, GB, SVM, SHAP, LIME	Accuracy, precision, recall, F1-score, ROC-AUC	Ensemble methods achieved the best predictive performance, with AUC values above 0.85.
Al-Makhadmeh and Tolba (IEEE Access, 2020)	Hybrid CNN–LSTM	Accuracy, sensitivity, specificity, ROC-AUC	The hybrid deep learning model outperformed traditional classifiers, with reported accuracy above 90% and AUC values above 0.90.
Muhammad et al. (IEEE Access, 2023)	KNN, RF, LR, SVM, GNB, DT	Accuracy, precision, recall, F1-score, AUC	KNN achieved the best performance, with 91.8% accuracy, 92.5% precision, 91.4% recall, 91.9% F1-score, and AUC of 90.27%.
Al Gharib et al. (IEEE Access, 2025)	RF, GB, SVM, XGBoost meta-learner, SHAP, LIME	Accuracy, F1-score, AUC, Cohen's Kappa, MCC	The hybrid ensemble achieved accuracy values between 88% and 91%, F1-scores between 0.88 and 0.91, and AUC values up to 0.96.

TABLE II
DATA SPLIT

Set	Purpose	Percentage
Train	Model fitting and hyperparameter search	80%
Validation	Model selection and threshold tuning	10%
Test	Final unbiased evaluation	10%

TABLE III
PREPROCESSING OPERATIONS APPLIED TO THE DATASET

Data component	Preprocessing operation
Patient identifier	Removed because it does not provide predictive information.
Age	Converted from days to years.
Height, weight, blood pressure	Unrealistic values were removed using predefined plausible ranges.
BMI	Engineered from height and weight and encoded as an ordinal variable.
Numerical variables	Standardized when required by the model.
Ordinal variables	Encoded while preserving their natural order.
Binary variables	Kept as binary indicators.
Target variable	Preserved as a binary classification label.

D. Hyperparameter Search Space

The following section presents the models trained in this study, the hyperparameters that were identified or explored, and the metrics selected to evaluate them.

A variety of models were evaluated to ensure a certain level of rigor, including a Parametric model, a Non-Parametric model, a Random Forest, a Support Vector Machine, and an Artificial Neural Network, in this case, a Multilayer Perceptron.

This configuration ensured that all models were selected using the same official metric, ROC-AUC, while additional metrics were retained for clinical interpretation.

Table VI reports approximate 95% confidence intervals for cross-validation ROC-AUC, estimated as $\bar{x} \pm 1.96 s/\sqrt{k}$, where k is the number of folds.

E. Model Results

Table IV summarizes the validation and test performance of the five evaluated classifiers. The models are ordered according to validation ROC-AUC, which was the official model-selection criterion. MLP obtained the highest validation ROC-AUC, followed very closely by Random Forest. The difference between both models was small, indicating comparable global discriminative ability.

Although Random Forest achieved higher accuracy and precision on the test set, its recall and F2-score were substantially lower than those of MLP, SVM, and KNN. This behavior indicates that Random Forest was more conservative when predicting positive cardiovascular disease cases. In contrast, MLP, KNN, and SVM prioritized sensitivity after threshold adjustment, reducing false negatives at the cost of increasing false positives.

Since false negatives are clinically relevant in cardiovascular disease screening, ROC-AUC was used as the official model-selection criterion, while recall, F2-score, false negatives, and false positives were analyzed as secondary operational criteria. Based on validation ROC-AUC, MLP was selected as the final model.

Fig. 1 compares the validation and test ROC-AUC values of the five evaluated models. MLP achieved the highest validation ROC-AUC, followed closely by Random Forest, indicating comparable global discriminative ability.

F. Threshold Adjustment

ROC-AUC was used as the official model-selection metric because it evaluates the ranking ability of the classifier independently of a fixed decision threshold. However, deploying a binary classifier requires converting model scores into class labels. Therefore, after selecting MLP as the final model according to validation ROC-AUC, its decision threshold was adjusted on the validation set.

The threshold was selected by maximizing the F2-score, which assigns more importance to recall than to precision. This criterion was chosen because, in cardiovascular disease screening, a false negative represents a patient with disease classified as healthy, which is clinically more critical than

TABLE IV
VALIDATION AND TEST PERFORMANCE OF THE EVALUATED MODELS

Model	ROC-AUC Val.	ROC-AUC Test	Acc. Test	Prec. Test	Recall Test	F2 Test	FN Test	FP Test
MLP	0.8067	0.8044	0.5997	0.5554	0.9585	0.8370	141	2605
Random Forest	0.8065	0.8036	0.7368	0.7699	0.6680	0.6862	1127	678
SVM	0.8004	0.7957	0.5391	0.5182	0.9817	0.8328	62	3099
KNN	0.7995	0.7985	0.5798	0.5425	0.9647	0.8347	120	2762
Logistic Regression	0.7966	0.7925	0.7317	0.7656	0.6601	0.6862	1127	678

TABLE V
HYPERPARAMETER SEARCH CONFIGURATION

Model	Hyperparameters explored	Selection metric
Logistic Regression	`C` $\in$ $\{0.001, 0.01, 0.1, 1, 10, 100\}$, `penalty` $\in$ {`l1`,`l2`}, `solver` $\in$ {`liblinear`,`saga`}, and `class_weight` $\in$ {`None`,`balanced`} were tuned using randomized search.	ROC-AUC
KNN	`n_neighbors` $\in$ $\{11, 21, 31, 51, 75, 101, 151, 201\}$, `weights` $\in$ {`uniform`,`distance`}, `metric=minkowski`, and `p` $\in$ $\{1, 2\}$ were tuned using randomized search.	ROC-AUC
Random Forest	`n_estimators` $\in$ $\{100, 200, 300, 500\}$, `max_depth` $\in$ {`None`, 10, 20, 30}, `min_samples_split` $\in$ $\{2, 5, 10\}$, `min_samples_leaf` $\in$ $\{1, 2, 4\}$, `max_features` $\in$ {`sqrt`,`log2`}, and `class_weight` $\in$ {`None`,`balanced`} were tuned using randomized search.	ROC-AUC
SVM	Linear and RBF variants were evaluated. For Linear SVM, `C` and `class_weight` were tuned. For RBF SVM, `C`, γ, and `class_weight` were tuned using an initial randomized search followed by a refined randomized search.	ROC-AUC
MLP	`hidden_layer_sizes`, `activation`, α, `learning_rate_init`, and `batch_size` were tuned using randomized search with early stopping.	ROC-AUC

TABLE VI
APPROXIMATE CROSS-VALIDATION ROC-AUC CONFIDENCE INTERVALS

Model	Mean CV ROC-AUC	Approx. 95% CI
Logistic Regression	0.7893	[0.7862, 0.7924]
KNN	0.7920	[0.7905, 0.7934]
Random Forest	0.7979	[0.7954, 0.8004]
SVM	0.7918	[0.7889, 0.7948]
MLP	0.7987	[0.7959, 0.8015]

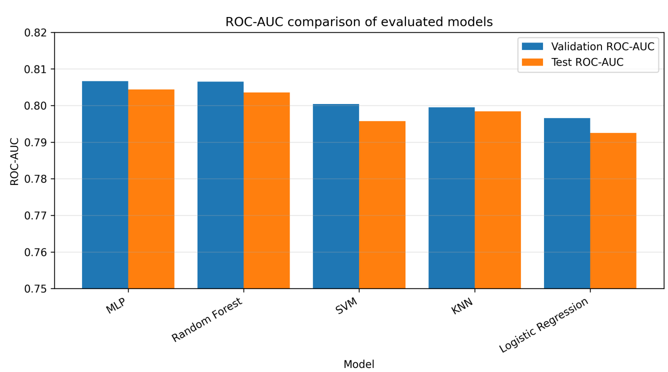

Fig. 1. Validation and test ROC-AUC comparison of the five evaluated models. Higher values indicate stronger global discriminative ability.

a false positive. The selected threshold was then fixed and applied once to the test set.

Fig. 2 shows the effect of the decision threshold on precision, recall, and F2-score for the final MLP model. Lower thresholds increase recall by classifying more patients as positive, whereas higher thresholds tend to increase precision but may increase the number of false negatives.

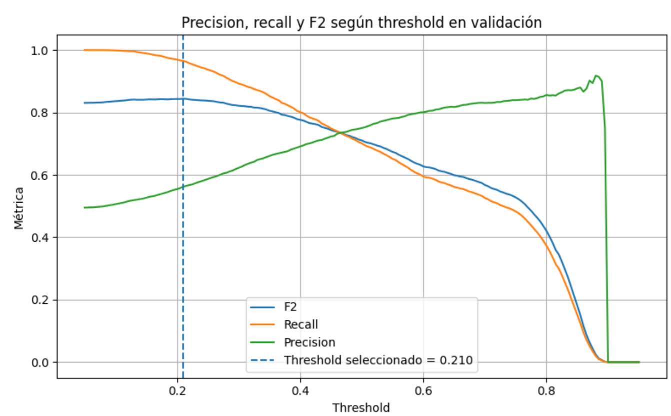

Fig. 2. Precision, recall, and F2-score as a function of the decision threshold on the validation set for the final MLP model. The selected threshold maximizes F2-score, prioritizing the detection of positive cardiovascular disease cases.

Fig. 3 presents the normalized confusion matrix of the final MLP model on the test set using the adjusted threshold. Percentages are computed row-wise with respect to the true class.

The adjusted threshold substantially increased recall and reduced false negatives, but also increased false positives. Therefore, the resulting classifier should be interpreted as a screening-oriented model rather than a definitive diagnostic system.

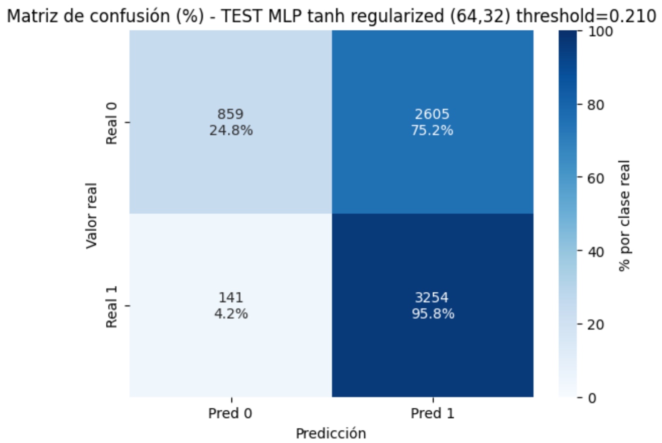

Fig. 3. Normalized confusion matrix of the final MLP model on the test set using the adjusted threshold. The matrix shows both absolute counts and row-wise percentages.

V. Dimensionality Reduction

A. Individual Feature Analysis

An individual feature analysis was performed using Mutual Information to estimate the discriminative contribution of each variable with respect to the target. Fig. 4 shows that systolic blood pressure, diastolic blood pressure, age, cholesterol, and BMI were the most informative variables. Lifestyle and demographic variables such as smoking, alcohol intake, height, and gender showed lower individual contribution. However, these variables were not automatically removed because this analysis is univariate and does not capture interactions between predictors.

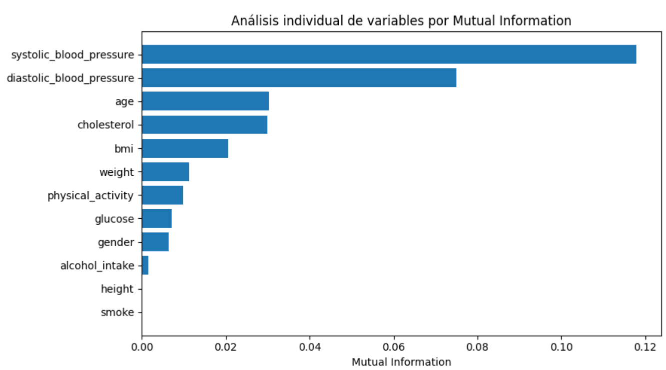

Fig. 4. Individual feature relevance estimated using Mutual Information. Higher values indicate stronger individual association with the cardiovascular disease target.

B. Linear Feature Extraction Using PCA

PCA was applied to the preprocessed representation of the dataset to evaluate whether dimensionality could be reduced while preserving predictive performance. The original preprocessed representation contained 16 dimensions. The number of principal components was selected using a cumulative explained variance criterion of at least 95%. As shown in Fig. 5, 9 components preserved approximately 96.99% of the variance, corresponding to a dimensionality reduction of 43.75%.

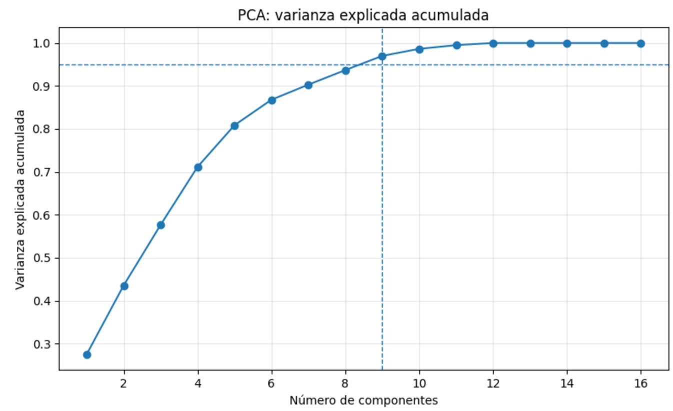

Fig. 5. Cumulative explained variance as a function of the number of PCA components. The selected configuration uses 9 components, preserving approximately 96.99% of the variance.

C. Nonlinear Feature Extraction Using UMAP

UMAP was evaluated as a nonlinear dimensionality reduction method using 2 components. This configuration reduced the dimensionality from 16 to 2 components, corresponding to an 87.5% reduction. Fig. 6 shows the two-dimensional UMAP representation of the training set. The visualization shows considerable overlap between the two classes, suggesting that the target classes are not cleanly separable in the two-dimensional UMAP space.

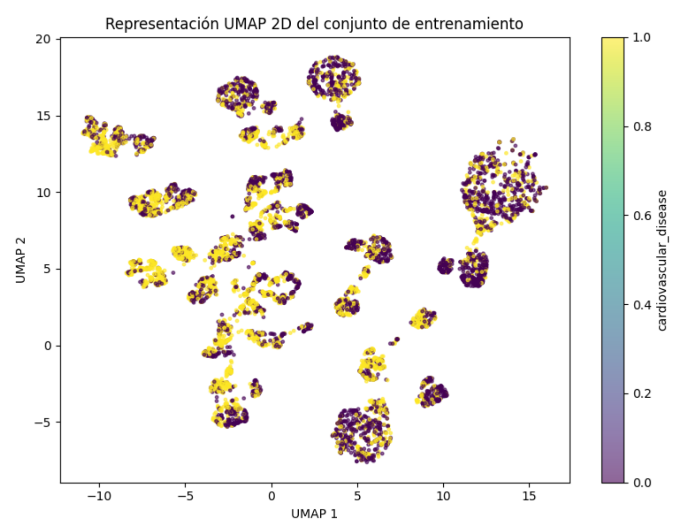

Fig. 6. Two-dimensional UMAP representation of the training set. Points are colored according to the cardiovascular disease target.

Table VII summarizes the performance of the two best models, MLP and Random Forest, using the original preprocessed representation, PCA, and UMAP. PCA preserved almost the same ROC-AUC as the original representation, whereas UMAP produced a clear degradation in ROC-AUC for both models.

TABLE VII
PERFORMANCE OF THE TOP TWO MODELS USING ORIGINAL, PCA, AND UMAP REPRESENTATIONS

Representation	Model	Components	Reduction	ROC-AUC Val.	ROC-AUC Test	F2 Test	FN / FP Test
Original	MLP	16	0.00%	0.8067	0.8044	0.8372	138 / 2615
Original	Random Forest	16	0.00%	0.8064	0.8036	0.8341	147 / 2641
PCA	MLP	9	43.75%	0.8054	0.8026	0.8381	94 / 2813
PCA	Random Forest	9	43.75%	0.8017	0.8003	0.8351	142 / 2643
UMAP	Random Forest	2	87.50%	0.7704	0.7689	0.8332	67 / 3063
UMAP	MLP	2	87.50%	0.7519	0.7546	0.8331	62 / 3091

Fig. 7 compares the ROC curves on the test set for the evaluated representations. The original and PCA representations showed similar discriminative behavior, while UMAP consistently reduced the ROC-AUC.

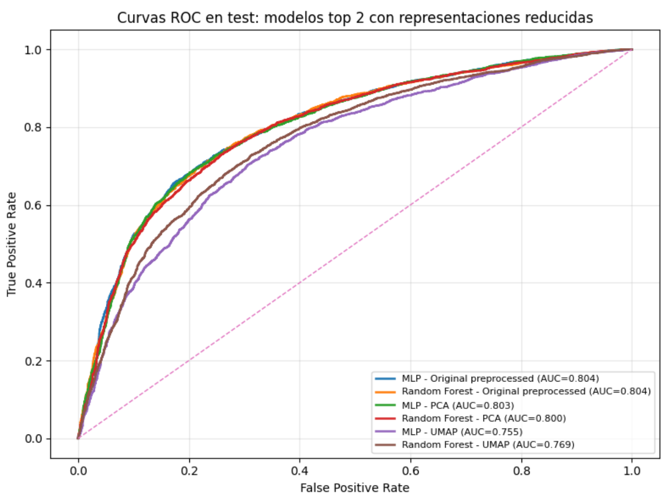

Fig. 7. ROC curves on the test set for the top two models using original, PCA, and UMAP representations.

VI. DISCUSSION

The experimental results show that MLP achieved the highest validation ROC-AUC among the five evaluated models, although Random Forest obtained a very similar ROC-AUC. This small difference suggests that both models have comparable global discriminative ability. However, their operational behavior differs substantially. Random Forest achieved higher test accuracy and precision, but lower recall and F2-score, indicating a more conservative classification behavior. In contrast, MLP, KNN, and SVM achieved much higher recall after threshold adjustment, which is desirable in a cardiovascular screening scenario where false negatives are more critical.

The selected MLP model should therefore be interpreted as a screening-oriented classifier rather than a definitive diagnostic system. Its adjusted threshold favors sensitivity, reducing the probability of missing positive cardiovascular disease cases, but this comes at the cost of increasing false positives. In practice, this behavior is acceptable only if positive predictions are followed by medical evaluation or confirmatory testing.

Dimensionality reduction showed that PCA is a reasonable alternative when model compactness is important. PCA reduced the representation from 16 to 9 components while preserving almost the same ROC-AUC. UMAP achieved a stronger reduction to 2 components, but degraded ROC-AUC substantially. Therefore, UMAP was useful mainly as an exploratory visualization tool, not as the final predictive representation.

Compared with the state-of-the-art studies reviewed in Section III, the ROC-AUC values obtained in this work are lower than some reported results above 0.85 or 0.90. This difference can be explained by differences in datasets, preprocessing criteria, validation protocols, model complexity, and feature availability. Nevertheless, the present work uses a consistent train/validation/test protocol and evaluates multiple model families under the same selection criterion, making the comparison among the implemented models internally coherent.

VII. CONCLUSION

This project developed and evaluated a supervised machine learning pipeline for cardiovascular disease prediction. Five model families were compared: Logistic Regression, KNN, Random Forest, SVM, and MLP. ROC-AUC was used as the official selection metric, while recall, F2-score, false negatives, and false positives were analyzed due to the clinical relevance of missed cardiovascular disease cases.

MLP was selected as the final model because it obtained the highest validation ROC-AUC. After threshold adjustment, the model prioritized recall and F2-score, making it more suitable for screening than for definitive diagnosis. The dimensionality reduction experiments showed that PCA can reduce the input representation from 16 to 9 components with minimal loss in predictive performance, whereas UMAP produced a stronger reduction but degraded ROC-AUC. Therefore, the original preprocessed representation with MLP remains the final selected configuration, while PCA is a viable alternative when dimensionality reduction is required.

Future work should include external validation on independent clinical datasets, calibration analysis, explainability techniques such as SHAP, and a more clinically informed threshold selection strategy.

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